DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE – RAIGAD -402 103

Semester Examination – May - 2019

Branch: Electrical Engineering **Sem.: IV**

Subject with Subject Code: Electromagnetic Theory [BTEEE406C]

Date: 24/05/2019 **Time: 3 Hrs.**

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

(Marks)

Q.1. A) A vector of magnitude 10 points from $(7, 2\pi/4, 0)$ in cylindrical coordinates towards the origin. Express the vector in Cartesian coordinates as well as spherical coordinates. **(06)**

B) Determine the divergence of following vector fields. **(06)**

$\mathbf{P} = yz\mathbf{a}_x + 4xy\mathbf{a}_y + 3y\mathbf{a}_z$ at (2,-3,4)

$\mathbf{Q} = \rho z \sin\phi\, \mathbf{a}_\rho + \rho z^2 \cos\phi\, \mathbf{a}_\phi$ at $(2, 2\pi/4, 1)$

Q.2. Answer any two

A) A point charge 1 mC and -2 mC are located at (3, 2, -1) and (1,-2,4), respectively. Calculate the electric force on a 10 nC charge located at (0, 3, 2) and the electric field intensity at that point. **(06)**

B) Derive an expression for the electric field intensity at an arbitrary point P (x, y, z), from a line charge. **(06)**

C) Determine D at (5, 3, 0) if there is a point charge -6π mC at (3,0,0) and line charge 3π mC along y axis. **(06)**

Q.3. A) State and prove Uniqueness theorem. **(08)**

OR

What is method of images? Explain with a point charge above grounded conducting plane.

B) Two extensive homogeneous isotropic dielectrics meet on plane z = 0. For z > 0, $\varepsilon_{r1} = 3$ and for z < 0, $\varepsilon_{r2} = 2$. A uniform electric field $E_1 = 5\mathbf{a}_x + -3\mathbf{a}_y + 3\mathbf{a}_z$ kV/m exists for $z \geq 0$ find E_2 for $Z \leq 0$. **(04)**

Q.4. A) Derive an expression for Biot Savart's Law and Ampere's Law **(12)**

A circular loop located on $x^2 + y^2 = 7$, z = 0 carries a direct current of 12 A along $\mathbf{a}_\phi$. Determine H at (0, 0, 3) and (0, 0, -3).

Q.5.	**A)**	**A parallel plate capacitor with plate area 6 cm^2 and plate separation 4 mm has a voltage $50\sin 10^3 t$ V applied to its plates. Calculate displacement current assuming $\varepsilon = \varepsilon_0$.**	**(04)**
	B)	**State and prove Poynting theorem.**	**(08)**
		OR	
	C)	**A uniform plane wave in air wi E=** $8\cos(\omega t - 3x - 4z)\mathbf{a_x}$ **V/m is incident on a dielectric slab ($z \geq 0$)with $\mu_r = 1$, $\varepsilon_r = 2.5$, σ = . Find (a) polarization of wave (b) angle of incidence (c) The reflected E field (d) The transmitted H field.**	**(08)**
Q.6.		**Derive an expressions for Propagation constant and characteristics equations of a transmission line. An air line has a characteristic impedance of 70Ω and phase constant of 4rad/m at 120 MHz. Calculate the inductance and capacitance per meter of the line.**	**(12)**

End